

1.3 Fourier-Motzkin Elimination for Cones

The main objective of this section is to prove Theorem 1.3. This again has two parts.

For the “forward direction” of Theorem 1.3, let $C = \text{cone}(Y) \subseteq \mathbb{R}^d$ be a $\mathcal{V}$ -cone. We can write it as

$$\begin{aligned} C &= \{Y\mathbf{t} \in \mathbb{R}^d : \mathbf{t} \geq \mathbf{0}\} \\ &= \{\mathbf{x} \in \mathbb{R}^d : \exists \mathbf{t} \in \mathbb{R}^n : \mathbf{t} \geq \mathbf{0}, \mathbf{x} = Y\mathbf{t}\}. \end{aligned}$$

The set $\{(\mathbf{x}, \mathbf{t}) \in \mathbb{R}^{d+n} : \mathbf{t} \geq \mathbf{0}, \mathbf{x} = Y\mathbf{t}\}$ clearly is an $\mathcal{H}$ -cone. Thus $C = \text{cone}(Y)$ can be written as the projection of this cone to the subspace $\{(\mathbf{x}, \mathbf{t}) \in \mathbb{R}^{d+n} : \mathbf{t} = \mathbf{0}\}$. Again, this projection can be formed successively, by projecting with respect to individual t_k -coordinates one by one. Thus it suffices to prove the following lemma.

Lemma 1.5. *If $C = P(A, \mathbf{0})$ is an $\mathcal{H}$ -cone in $\mathbb{R}^d$, then so is the elimination $\text{elim}_k(C) = \{\mathbf{x} - te_k : \mathbf{x} \in C, t \in \mathbb{R}\}$, and thus also the projection $\text{proj}_k(C) = \text{elim}_k(C) \cap H_k$. Namely, we get $\text{elim}_k(C) = P(A^{/k}, \mathbf{0})$ for*

$$A^{/k} := \{\mathbf{a}_i : a_{ik} = 0\} \cup \{a_{ik}\mathbf{a}_j + (-a_{jk})\mathbf{a}_i : a_{ik} > 0, a_{jk} < 0\}.$$

(Here we interpret A and $A^{/k}$ as sets of row vectors.)

Proof. The row vectors in $A^{/k}$ are positive combinations of row vectors in A , hence the corresponding inequalities are valid for C , and thus we get that $C \subseteq P(A^{/k}, \mathbf{0})$. Furthermore, the row vectors in $A^{/k}$ all have the k th component equal to zero (by construction), i.e., the variable x_k does not appear in the system $A^{/k}\mathbf{x} \leq \mathbf{0}$, which proves that $\text{elim}_k(C) \subseteq P(A^{/k}, \mathbf{0})$.

For the converse, let $\mathbf{x} \in P(A^{/k}, \mathbf{0})$, and let $x_k = 0$ (without loss of generality). We claim that $\mathbf{x} - ye_k \in C$ for suitable y . In fact, plugging $\mathbf{x} - ye_k$ into the system $A\mathbf{x} \leq \mathbf{0}$, we find that y has to satisfy

$$\max_i \left\{ \frac{1}{a_{ik}} \mathbf{a}_i \mathbf{x} : a_{ik} > 0 \right\} \leq y \leq \min_j \left\{ \frac{1}{-a_{jk}} (-\mathbf{a}_j) \mathbf{x} : a_{jk} < 0 \right\}.$$

This can be satisfied. Namely, if $a_{ik} > 0$ and $a_{jk} < 0$, then we know that $\frac{1}{a_{ik}} \mathbf{a}_i \mathbf{x} \leq \frac{1}{-a_{jk}} (-\mathbf{a}_j) \mathbf{x}$, which is equivalent to $(a_{ik}\mathbf{a}_j + (-a_{jk})\mathbf{a}_i) \mathbf{x} \leq 0$, which holds because $\mathbf{x} \in P(A^{/k}, \mathbf{0})$. $\square$

Now we proceed to prove the “backward direction” of Theorem 1.3. For this let $C = P(A, \mathbf{0}) \subseteq \mathbb{R}^d$ be an $\mathcal{H}$ -cone. We can write it as

$$\begin{aligned} C &= \{\mathbf{x} \in \mathbb{R}^d : A\mathbf{x} \leq \mathbf{0}\} \\ &\cong \left\{ \begin{pmatrix} \mathbf{x} \\ \mathbf{w} \end{pmatrix} \in \mathbb{R}^{d+m} : A\mathbf{x} \leq \mathbf{w} \right\} \cap \left\{ \begin{pmatrix} \mathbf{x} \\ \mathbf{w} \end{pmatrix} \in \mathbb{R}^{d+m} : \mathbf{w} = \mathbf{0} \right\}. \end{aligned}$$

Here $\{(\mathbf{x}) \in \mathbb{R}^{d+m} : A\mathbf{x} \leq \mathbf{w}\}$ is a $\mathcal{V}$ -cone, as we have shown above.

The intersection with $\{(\mathbf{x}) \in \mathbb{R}^{d+m} : \mathbf{w} = \mathbf{0}\}$ can be formed successively, by setting coordinates to zero one at a time, i.e., intersecting with coordinate hyperplanes of the form $H_k = \{\mathbf{y} \in \mathbb{R}^{d+m} : y_k = 0\}$. Thus it suffices to prove the following lemma.

Lemma 1.6. *If $C = \text{cone}(Y)$ is a $\mathcal{V}$ -cone in $\mathbb{R}^d$, then so is the intersection $C \cap H_k$. Namely, we get $C \cap H_k = \text{cone}(Y^{/k})$ for*

$$Y^{/k} := \{\mathbf{y}_i : y_{ki} = 0\} \cup \{y_{ki}\mathbf{y}_j + (-y_{kj})\mathbf{y}_i : y_{ki} > 0, y_{kj} < 0\}.$$

(Here we interpret Y and $Y^{/k}$ as sets of column vectors. Thus y_{ki} denotes the k th component of $\mathbf{y}_k$, and thus the (k, i) -entry of the matrix $Y = (\mathbf{y}_1, \dots, \mathbf{y}_n)$, accordance with the notation introduced on page 2.)

Proof. First note that the vectors in $Y^{/k}$ all have x_k -coordinate 0, so clearly $C \cap H_k \supseteq \text{cone}(Y^{/k})$.

For the reverse inclusion, we consider some $\mathbf{v} = Y\mathbf{t} \in \text{cone}(Y)$ ($\mathbf{t} \geq 0$) with $v_k = 0$. Now either we have $t_i y_{ki} = 0$ for all i , in which case we get $\mathbf{v} \in \text{cone}(\{\mathbf{y}_i : y_{ki} = 0\})$, or we can expand $v_k = 0$, to get

$$\Lambda := \sum_{i: y_{ki} > 0} t_i y_{ki} = \sum_{j: y_{kj} < 0} t_j (-y_{kj}) > 0.$$

With this, we can rewrite $\mathbf{v}$ as

$$\begin{aligned} \mathbf{v} &= \sum_{i: y_{ki} = 0} t_i \mathbf{y}_i + \sum_{i: y_{ki} > 0} t_i \mathbf{y}_i + \sum_{j: y_{kj} < 0} t_j \mathbf{y}_j \\ &= \sum_{i: y_{ki} = 0} t_i \mathbf{y}_i + \frac{1}{\Lambda} \sum_{i: y_{ki} > 0} \left(\sum_{j: y_{kj} < 0} t_j (-y_{kj}) \right) t_i \mathbf{y}_i \\ &\quad + \frac{1}{\Lambda} \sum_{j: y_{kj} < 0} \left(\sum_{i: y_{ki} > 0} t_i y_{ki} \right) t_j \mathbf{y}_j \\ &= \sum_{i: y_{ki} = 0} t_i \mathbf{y}_i + \sum_{\substack{i: y_{ki} > 0 \\ j: y_{kj} < 0}} \frac{t_i t_j}{\Lambda} ((-y_{kj})\mathbf{y}_i + y_{ki}\mathbf{y}_j). \end{aligned}$$

This proves the claim, by giving an *explicit* representation of $\mathbf{v}$ as a conical sum of vectors in $Y^{/k}$. $\square$

This completes the proof of Theorem 1.3. In this proof, we have provided explicit projections and intersections — one could also argue less explicitly that coefficients exist.

For example, in the last proof the space of possible coefficients turns out to be a transportation polytope, of which we have implicitly used the special interior point; see Problem 1.8. An alternative approach is to use a vertex, and thus to get fewer nonzero coefficients.

How efficient is Fourier-Motzkin elimination as a computational tool? The main problem is that the number of inequalities generated by this method can go beyond all tractable limits within a few elimination steps. For this observe that if A has m rows, then $A^{/k}$ may have as many as $\lfloor \frac{m^2}{4} \rfloor$ rows: the number of inequalities can roughly be squared by every step, which leads to problems even with a fast implementation on a computer with a lot of memory.

Nevertheless the computations can be carried out fairly efficiently. See the notes to this lecture for comments about the available programs.

Let us mention only that Fourier-Motzkin elimination can in principle be used as an algorithm for linear programming (see Section 3.2). In fact, to find a point in $P(A, z)$ which maximizes $\mathbf{c}\mathbf{x}$, we introduce an extra variable $x_0 = \mathbf{c}\mathbf{x}$, and eliminate all the other variables. This will tell us the possible range of x_0 , and by backtracking one can recover the optimal basis (i.e., a set of inequalities whose nonnegative combination yields an optimal upper bound on x_0). However, this method for linear programming is exponential, and there are much better ones available.

1.4 The Farkas Lemma

It was first pointed out by Kuhn [345] that with (termination of) Fourier-Motzkin elimination we have also done all the work for the *Farkas lemma*. This extremely important lemma appears in many different versions all over the theory of polytopes and polyhedra. It is interesting to note that if you look into different books and papers, you find quite different lemmas all called “the Farkas lemma.” All of these, however, are easily transformed into each other.

Essentially, the Farkas lemma yields a characterization for the solvability of a system of inequalities. There are variations for systems of inequalities in various standard forms: Farkas lemmas for polyhedra and for cones, for inequality systems with equalities, inequalities or strict inequalities, in nonnegative, positive or unrestricted variables, and so on. There are also quite different ways to formulate theorems “of Farkas type”:

- as *theorems of the alternative* (one inequality system has a solution if and only if a second system has none),
- as *transposition theorems* (because the second system can be derived by transposing the matrix and vectors of the first),
- as *duality theorems* (the duality theorem for linear programs is of Farkas type),
- as *good characterizations* (if a system has a solution, then any solution vector proves this; if it has no solution, then the Farkas lemma yields a dual vector that encodes this fact),